Michel Rouly North America **web** michel.rouly.net **github** jrouly **gpg** 0x1DFE3B2C

Technical Skills

Languages	Scala, Python, TypeScript, Java
Data Platforms	Apache Iceberg, Apache Spark, Dagster, PostgreSQL, Elasticsearch, Redis
Infrastructure	AWS, Kubernetes, Docker, Terraform, Helm, CDK, Prometheus, Grafana

Work Experience

ZEPHYR AI {python, machine learning, infrastructure} zephyrai.bio
Software Architect **04/2021 - present**

- Architected and delivered a unified data and AI platform supporting secure, hybrid-cloud data pipelines and ML workflows in a regulated environment (HIPAA, SOC2), accelerating time-to-production for new models and data sources.
- Implemented a lakehouse architecture (Apache Iceberg on S3 with AWS Glue catalog), supporting large-scale analytic and ML workloads with strong auditability guarantees and multi-engine interoperability (Spark, Python), enabling consistent access patterns across batch and exploratory workflows.
- Developed a distributed orchestration layer (Dagster with Python and Spark) on a standardized and cost optimized compute substrate (EKS), managing end-to-end data and ML pipelines including ingestion, feature engineering, model training, and batch inference, improving pipeline reliability and reducing operational overhead for internal customers.
- Defined architecture for cross-cloud (AWS/Azure) data and ML workflows, addressing data locality, compliance boundaries, and access control, enabling execution of pipelines across environments.
- Established data governance and quality standards, including dataset versioning, lineage tracking, and validation standards, reducing compliance risk while enabling reproducible ML experimentation and collaboration across teams.
- Introduced platform-wide observability (Prometheus, Grafana, central logging, alerting), improving platform reliability, enabling proactive incident response, and reducing time to resolution for pipeline and model failures.
- Led architectural direction and technical strategy for the platform, aligning infrastructure, data, and ML systems with long-term scalability, cost efficiency, and regulatory constraints; drove cross-functional alignment across engineering, data science, research science, and external stakeholders.
- Optimized cloud infrastructure costs and improved system reliability through storage tiering (S3 tiering and lifecycle policies), compute strategies (spot market), and data format standardization (Parquet).

RALLY HEALTH {scala, microservices, infrastructure} rallyhealth.com
Principal Software Engineer **11/2019 - 04/2021**
Senior Software Engineer **09/2016 - 11/2019**
Software Engineer **09/2015 - 09/2016**

- Developed greenfield user facing microservice based applications.
- Managed production deploys; served on-call rotations; supported 24/7 live site.
- Designed and optimized ETL pipelines using Akka Streams; accelerated data velocity and increased data quality standards.
- Built internal developer tooling including sbt plugins, focusing on automation.
- Instrumented performance and availability metrics with Datadog; increased application observability and reliability.
- Implemented security engineering goals with a focus on user privacy.

GEORGE MASON UNIVERSITY {python, django, open source} honorscollege.gmu.edu
Lead Project Developer **08/2013 - 01/2015**

HARMONIA {java, machine learning} harmonia.com
Programming Consultant **06/2012 - 06/2013**

Education

M.S. Computer Science, GEORGIA INSTITUTE OF TECHNOLOGY **01/2017 - 12/2018**
B.S. Computer Science, GEORGE MASON UNIVERSITY, *summa cum laude* **08/2011 - 06/2015**

Publications

2015 ACM doi.org/10.1145/2787622.2787723
2014 ACM doi.org/10.1145/2688204.2688207
2014 IEEE doi.org/10.1109/IC2E.2014.42

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